UNIVERSITY OF MAURITIUS
SIS 3204Y(5) – NATURAL LANGUAGE PROCESSING

MOCK PAPER 1 – Weeks 1 to 8 (Question Paper with Model Answers)
Total: 50 marks | Suggested duration: 60 minutes

Instructions: Answer ALL questions. Marks are shown beside each question. The week each question comes from is shown in the question header. Model answers are given under each question in bullet points; the mark for a point is shown in brackets, e.g. (1). Cover the answers while practising. Equivalent correct answers should receive credit.

Question 1 | Week 1 (Revision Q3) [6 marks]

Explain why human language is difficult for computers to understand.

Answer

- **Ambiguity:** the same word or sentence can have many meanings, e.g. “bank” can be a financial institution or the side of a river. (1)
- **Context dependence:** meaning depends on the situation and the surrounding words. (1)
- **Sarcasm and irony:** the words may mean the opposite of their literal meaning. (1)
- **Idioms and figurative language:** expressions like “break a leg” cannot be understood literally. (1)
- **Language variations:** slang, abbreviations, typos and regional dialects. (1)
- **Emotions and cultural differences:** tone, feelings and background knowledge change the interpretation. (1)

Question 2 | Week 2 (Revision Q18) [6 marks]

Compare stemming and lemmatisation in a table using FOUR criteria. Give examples for both.

Answer

- **Method:** stemming uses simple rules to cut prefixes/suffixes; lemmatisation uses vocabulary and morphological analysis with POS and context. (1)
- **Output:** stemming may give non-words (studies → studi); lemmatisation gives valid dictionary words (studies → study). (1)
- **Speed:** stemming is faster and mostly language independent; lemmatisation is slightly slower and language dependent. (1)
- **Accuracy:** stemming is less accurate and may lose meaning; lemmatisation is more accurate and keeps meaning and context. (1)
- **Stemming examples:** running → run, happily → happi. (1)
- **Lemmatisation examples:** mice → mouse, went → go. (1)

Question 3 | Week 3 (Revision Q4) [5 marks]

Define a vocabulary. Given the corpus: Doc 1 = “the cat sat on the mat” and Doc 2 = “the dog sat on the log”, build the vocabulary (assign indices from 0 in order of first appearance) and integer-encode Doc 2.

Answer

- A vocabulary is the list of all **unique tokens** (words) in the corpus; each token gets a unique index used later for encoding. (1)
- Vocabulary (|V| = 7): the=0, cat=1, sat=2, on=3, mat=4, dog=5, log=6. (2)
- Doc 2 tokens: [the, dog, sat, on, the, log]. (1)
- Integer-encoded Doc 2 = **[0, 5, 2, 3, 0, 6]** (both “the” use index 0). (1)

Question 4 | Week 4 (Part B Q2) [4 marks]

Explain how a neural network learns from an incorrect prediction: Prediction → Loss → Backpropagation → Gradient Descent → Weight Update.

Answer

- The **prediction** is compared with the true label and the **loss** is calculated. (1)
- **Backpropagation** sends the error backward through the network and computes the gradients. (1)
- **Gradient descent** uses the gradients and the learning rate to decide how to change the parameters. (1)
- Weights and biases are **updated** so that the loss is smaller in future predictions. (1)

Question 5 | Week 5 (Sample Q4) [8 marks]

Explain the role of the forget gate, input gate and output gate in an LSTM. How do these gates help an LSTM process long sequences?

Answer

- **Forget gate:** decides which information from the previous cell state is kept or discarded (close to 0 = forget, close to 1 = keep). (2)
- **Input gate:** decides which new information is added to the cell state. (2)
- **Output gate:** decides which part of the updated cell state goes into the current hidden state. (2)
- Together the gates give **selective memory**: useful information is remembered and irrelevant information is removed. (1)
- With the cell state, this keeps important information over long sequences and reduces the long-term dependency (vanishing gradient) problem of a standard RNN. (1)

Question 6 | Week 6 (Set 1 Q6) [10 marks]

Write the scaled dot-product attention formula and explain the purpose of each part of the formula.

Answer

- **Formula:** Attention(Q, K, V) = softmax(QK^T / $\sqrt{d_k}$) V (2)
- **QK^T:** dot product of Queries and Keys; gives relevance (similarity) scores. (2)
- **Division by $\sqrt{d_k}$:** stops the scores from becoming too large when the Key dimension is large; keeps softmax stable. (2)
- **Softmax:** converts the scaled scores into attention weights (positive values that sum to 1). (2)
- **Multiply by V:** the weights are applied to the Value vectors to give the contextualised output. (2)

Question 7 | Week 7 (Set 1 Q3) [5 marks]

Consider the sentence: “The customer was satisfied with the service.”

a) Identify the Transformer architecture used by BERT. (1 mark)

b) Explain how BERT uses bidirectional context to represent the word “satisfied”. (2 marks)

c) Describe how BERT could be adapted to classify the sentiment of this sentence. (2 marks)

Answer

(a)

- BERT uses an **encoder-only** Transformer. (1)

(b)

- Bidirectional self-attention looks at both the **left and right** words around “satisfied”. (1)
- So the representation of “satisfied” is influenced by the **whole sentence**. (1)

(c)

- Add a **classification head**, commonly on the [CLS] representation. (1)
- **Fine-tune** on labelled sentiment examples so it predicts the sentiment. (1)

Question 8 | Week 8 (Tutorial Set A Q3) [5 marks]

A university wants to classify student feedback as Positive, Negative or Neutral. The initial prompt is: “Analyse this feedback: The lecturer explains well, but the laboratory computers are frequently unavailable.”

(a) Identify two weaknesses of the initial prompt. (2 marks)

(b) Rewrite the prompt to make the task clearer and output more consistent. (3 marks)

Answer

(a)

- It does not state the **allowed labels** (Positive, Negative, Neutral). (1)
- It does not explain how **mixed sentiment** should be classified. (1)

(b) Example improved prompt

- “Classify the student feedback as exactly one of: Positive, Negative or Neutral. If both positive and negative views appear, use Neutral. Return only the label. Feedback: ‘The lecturer explains well, but the laboratory computers are frequently unavailable.’” Expected output: **Neutral**.
- Task and labels clearly stated. (1)
- Rule for mixed sentiment. (1)
- Consistent output format (“return only the label”). (1)

Question 9 | Week 4 (Part A Q8) [1 mark]

Why is Softmax useful for multi-class text classification?

Answer

- It converts the output scores into **class probabilities that sum to 1**. (1)

Marks Summary

Question	Week	Source	Marks
1	1	Revision Q3	6
2	2	Revision Q18	6
3	3	Revision Q4	5
4	4	Part B Q2	4
5	5	Sample Q4	8
6	6	Set 1 Q6	10
7	7	Set 1 Q3	5
8	8	Tutorial Set A Q3	5
9	4	Part A Q8	1
		TOTAL	**50**